

- Correlation Analysis attempts to determine the degree of relationship between variables-**Ya-Kun-Chou**.
- Correlation is an analysis of the covariation between two or more variables.- **A.M.Tuttle**.

Correlation expresses the inter-dependence of two sets of variables upon each other. One variable may be called as (subject) independent and the other relative variable (dependent). Relative variable is measured in terms of subject.

Types of Correlation

Correlation is classified into various types. The most important ones are

- Positive and negative.
- Linear and non-linear.
- Partial and total.
- Simple and Multiple.

Positive Correlation

It depends upon the direction of change of the variables. If the two variables tend to move together in the same direction (ie) an increase in the value of one variable is accompanied by an increase in the value of the other, (or) a decrease in the value of one variable is accompanied by a decrease in the value of other, then the correlation is called **positive or direct correlation**.

Examples: Price and supply, height and weight, yield and rainfall.

Negative Correlation

If the two variables tend to move together in opposite directions so that increase (or) decrease in the value of one variable is accompanied by a decrease or increase in the value of the other variable, then the correlation is called **negative (or) inverse correlation**.

Examples: Price and demand, yield of crop and price.

Linear and Non-linear correlation

If the ratio of change between the two variables is a constant then there will be **linear correlation** between them. Consider the following.

X	2	4	6	8	10	12
Y	3	6	9	12	15	18

Here the ratio of change between the two variables is the same. If we plot these points on a graph we get a straight line.

If the amount of change in one variable does not bear a constant ratio of the amount of change in the other. Then the relation is called **Curvi-linear (or) non-linear correlation**. The graph will be a curve.

Simple and Multiple correlation

When we study only two variables, the relationship is simple correlation.

For example, quantity of money and price level, demand and price.

But in a multiple correlation we study more than two variables simultaneously.

The relationship of price, demand and supply of a commodity are an example for multiple correlation.

Partial and total correlation

The study of two variables excluding some other variable is called **Partial correlation**.

For example, we study price and demand eliminating supply side.

In total correlation all facts are taken into account.

- Scatter Diagram
- Graphical method
- Karl Pearson's Method
- Spearman's Rank Correlation Method
- Concurrent deviation Method
- Method of Least Squares

Karl Pearson Co-efficient Correlation:-

The measure for relationship between two variables is called the measure of correlation (or) correlation co-efficient it is denoted by ρ .

Karl Pearson Coefficient of correlation $\rho = \frac{\text{cov}(X, Y)}{\sigma_x \cdot \sigma_y}$

where $\text{cov}(X, Y) = \frac{\sum xy}{n} - \bar{x} \bar{y}$,

$$\sigma_x = \sqrt{\frac{\sum x^2}{n} - \bar{x}^2} \text{ and } \sigma_y = \sqrt{\frac{\sum y^2}{n} - \bar{y}^2}$$

where $\bar{x}, \bar{y}$ are the mean for X, Y .

n is the number of pairs of observations.

[Note:- $\text{cov}(X, Y) = E(XY) - E(X)E(Y)$

$$\sigma_x = \sqrt{\text{var}(X)} = \sqrt{E(\tilde{X}) - [E(X)]^2}$$

$$\sigma_y = \sqrt{\text{var}(Y)} = \sqrt{E(Y^2) - [E(Y)]^2}$$

Remark:- 1) $-1 \leq \rho \leq 1$.

If $\rho = 1$, then it is a perfect positive correlation

If $\rho = -1$ then it is a perfect negative correlation.

If $\rho = 0$ then there is no relationship between the variables.

2) The correlation coefficient of X and Y is symmetric,

$$(i) \quad \rho_{xy} = \rho_{yx}$$

Prob 1 Find the correlation coefficient between annual advertising expenditures and annual sales revenue for the following data:

Year	1	2	3	4	5	6	7	8	9	10
Annual advertising expenditure (x_i)	10	12	14	16	18	20	22	24	26	28
Annual sales (y_i)	20	30	37	50	56	78	89	100	120	110

Soln:-

$$P = \frac{\text{cov}(X, Y)}{\sigma_X \sigma_Y} \quad \text{where } \text{cov}(X, Y) = \frac{\sum xy}{n} - \bar{x}\bar{y};$$

$$\sigma_X = \sqrt{\frac{\sum x^2}{n} - \bar{x}^2}; \quad \sigma_Y = \sqrt{\frac{\sum y^2}{n} - \bar{y}^2}$$

x	y	xy	x ²	y ²
10	20	200	100	400
12	30	360	144	900
14	37	518	196	1369
16	50	800	256	2500
18	56	1008	324	3136
20	78	1560	400	6084
22	89	1958	484	7921
24	100	2400	576	10000
26	120	3120	676	14400
28	110	3080	784	12100
190	690	15004	3940	58810

$$\bar{x} = \frac{190}{10} = 19$$

$$\bar{y} = \frac{690}{10} = 69$$

$$\bar{x}\bar{y} = 1311$$

$$\begin{aligned} \text{cov}(X, Y) &= \frac{\sum xy}{n} - \bar{x}\bar{y} \\ &= \frac{15004}{10} - 1311 \\ &= 189.4 \end{aligned}$$

$$\sigma_X = \sqrt{\frac{\sum x^2}{n} - \bar{x}^2} = \sqrt{\frac{3940}{10} - 361} = \sqrt{33}$$

$$\sigma_Y = \sqrt{\frac{\sum y^2}{n} - \bar{y}^2} = \sqrt{\frac{58810}{10} - 4761} = \sqrt{1120}$$

$$\therefore P = \frac{189.4}{\sqrt{33} \sqrt{1120}} = 0.98$$